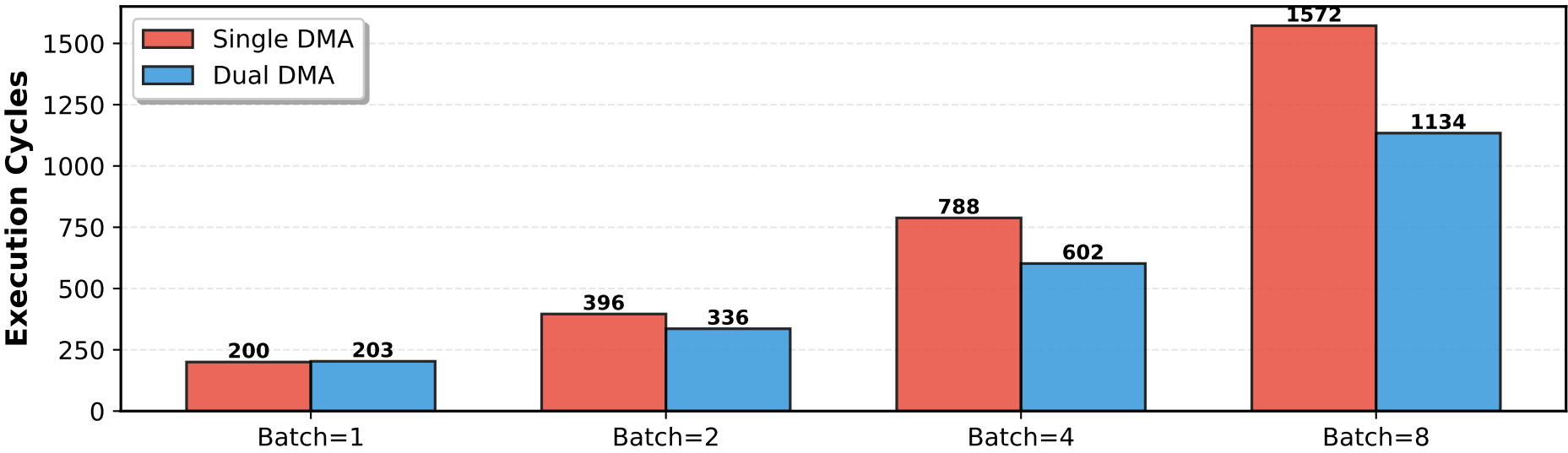
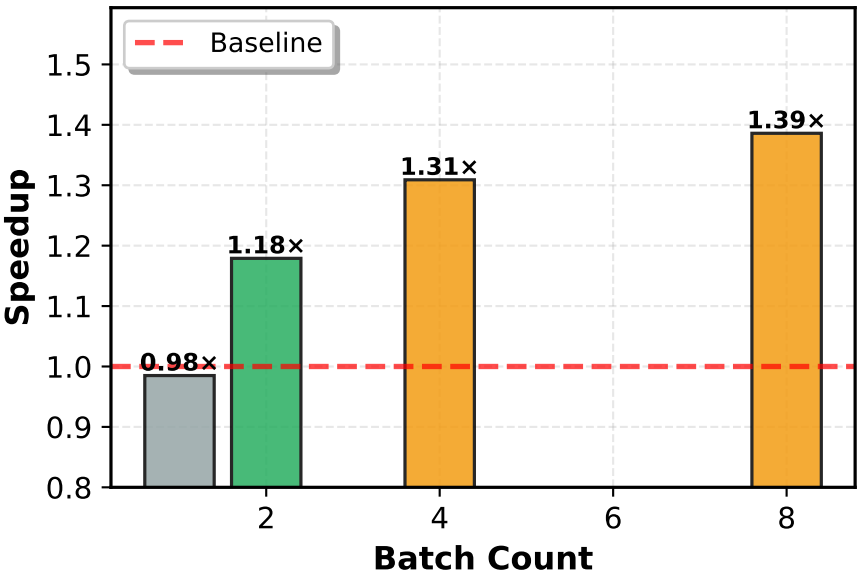


Single DMA vs Dual DMA: Comprehensive Performance Analysis

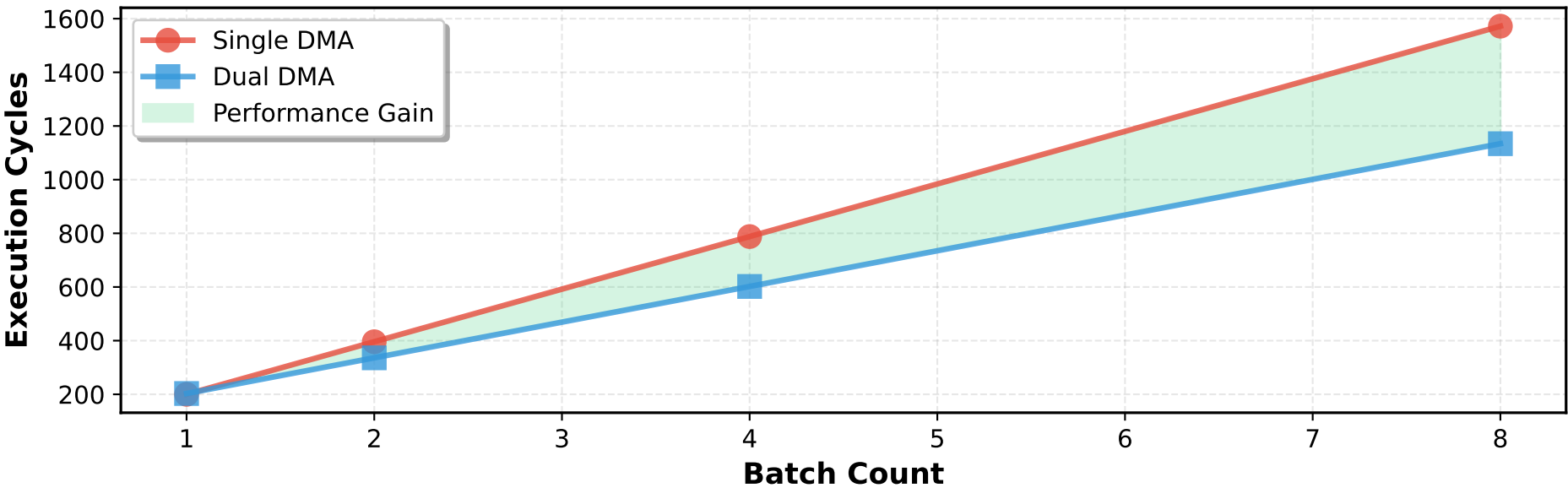
Absolute Execution Cycles Comparison



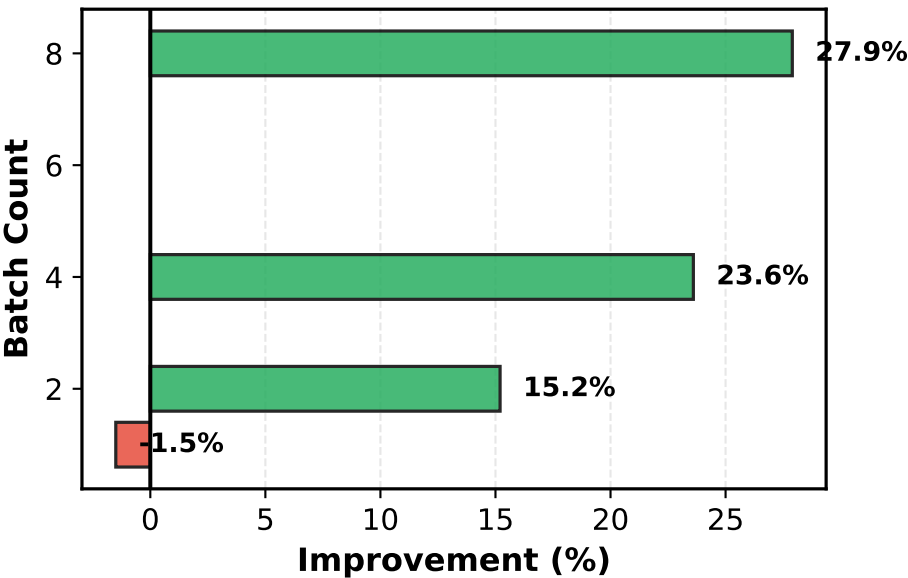
Dual DMA Speedup



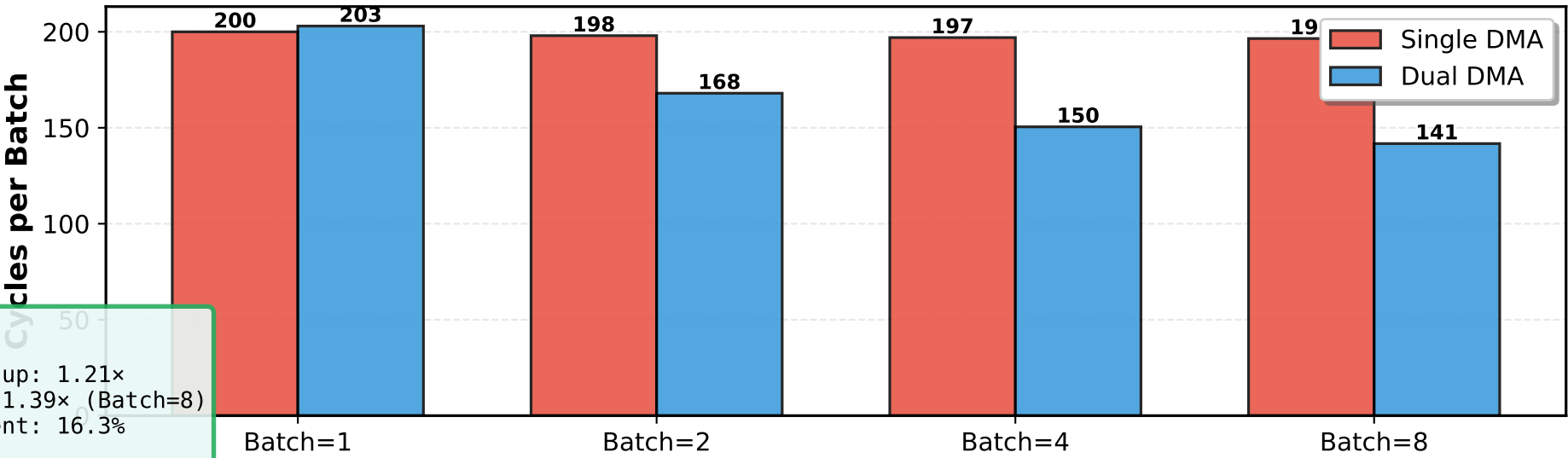
Performance Scaling with Batch Size



Performance Improvement



Efficiency: Average Cycles per Batch



Single DMA Architecture



Dual DMA Architecture



Key Findings:

- Average Speedup: 1.21x
- Max Speedup: 1.39x (Batch=8)
- Avg Improvement: 16.3%

Dual DMA Benefits:

- ✓ Independent Read/Write
- ✓ Better scaling with batches
- ✓ ~21% faster on average